Voxel Relay Game

Revised 29 October 2026. A single-player 3D voxel platform game: a repair robot restores a research facility by collecting energy cores and reaching the unlocked exit. The current prototype includes two timed missions and a guided, untimed training ground.

Gameplay and presentation

Default controls are camera-relative WASD movement, mouse orbit/wheel zoom, Space for variable-height jumps, Shift to sprint, C to toggle crouch or slide at speed, and E to collect cores, use relays and activate checkpoints. Crouching fits one-cell-high tunnels; momentum supports slide-jumps. Controls can be rebound and saved locally.

The courtyard introduces three cores, jumping and electric traps; the corridor mission combines five cores, narrow passages and moving hazards. Relays temporarily disable traps. Pressure plates open delayed-closing doors, while speed pads provide running boosts, sprint-triggered low flight or turbo slides. Temporary spring boots and a single-hit shield offer optional assistance. Training teaches these mechanics without a timer or life limit.

Each configuration has two checkpoints. Mission runs have three lives: destruction normally restores armour at the latest checkpoint (or entry), retains collected cores, and costs ten seconds; the final life restarts the level, while timeout ends the run. Checkpoints are in-run recovery, not disk saves. Black-screen transitions separate levels and full restarts.

A compact, articulated robot uses bouncy running, arm swings, crouch/slide poses and directional reactions. Blunt and electrical hits have distinct motion and effects; destruction launches the robot into a flat landing. Collection, impacts and movement combine particle bursts, brief hit-stop and camera shake. A clean facility palette, large bold typography, holographic HUD and interactive menu robot provide a readable science-fiction identity.

Technical scope and responsibilities

Technical baseline

C++17/OpenGL 3.3 Core, GLSL, a 64-cubed voxel grid, time-based movement and collision substeps support the current prototype. Textured surfaces use ambient, diffuse and specular lighting; parent-child transforms animate robot limbs and doors. Our two challenges remain normal mapping with an on/off comparison and event-driven particles with spawning, motion, lifetime and a reusable 1,024-particle pool. Internal-face removal is the separate core optimisation; measure it against the unculled baseline. Procedural death motion is not a full ragdoll simulation.

Planned ownership and integration

Jacky Hui: own movement, interaction, hazards, checkpoints/lives, training, power-ups, controls and menus; maintain robot animation, feedback and particle lifecycle. Jerry Xie: own voxel terrain, level layouts, textures, lighting, normal mapping, environment rendering and optimisation measurements; lead VLAB/CSE build validation. Together: agree interfaces, review integration, balance routes, verify resets and respawns, record asset sources and actual contributions, and prepare the report and live demonstration.

Scope control. Stabilise existing extensions before adding features; optional mechanics must not delay core requirements or the two challenge demonstrations.

Milestones and verification

Current baseline (29 October). Core loop, both missions, training and the extensions above are implemented. The current Windows checkout passes 57 automated tests, including checkpoint recovery, crouch clearance, life rules, training, power-ups and native input. These checks do not establish visual quality or VLAB compatibility.

Revised plan (by 1 November). Incorporate tutor feedback, confirm ownership and freeze scope; submit this one-page ProjectPlan.pdf in the repository root.

Final submission (by 15 November). Complete VLAB/CSE build and play-through checks; tune camera, collision, readability and difficulty; capture comparable optimisation timings/counts and normal-map/particle evidence. Finish the 3–6-page Report.pdf with accurate contributions and references, verify packaging, and rehearse the Week 10 live demonstration.